

- Quick Week 2 Module.

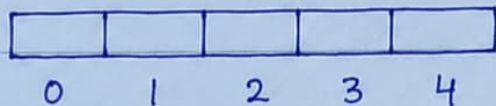
1. input $\rightarrow$ No. of id's we have. 5
 input $\rightarrow$ ID we are finding. 1450
 input $\rightarrow$ All 5 transaction ID.

o/p : fraud transaction found

OR

Transaction not found.

Linear search.



```
for ( int i = 0 ; i = 5 ; i++ )  
{
```

```
    if ( ID[i] == 1450 ) {
```

```
        cout << " Transaction found " ;
```

```
    }
```

```
}
```

```
}
```

```
#include <iostream>
```

```
#include <vector>
```

```
int main() {
```

```
    using namespace std;
```

```
    int NO_of_ID, LogTarget_ID;
```

```
    std::cout << "Enter no. of ID : ";
```

```
    std::cin >> NO_of_ID;
```

```
    std::vector<int> arr(NO_of_ID);
```

```
    std::cout << "Enter Log IDs : ";
```

```
    for (int i = 0; i < NO_of_ID; i++) {
```

```
        std::cin >> arr[i];
```

```
    }
```

```
    std::cout << "Enter target Log ID : ";
```

```
    std::cin >> LogTarget_ID;
```

```
    int start = 0;
```

```
    int end = NO_of_ID - 1;
```

```
    bool found = false;
```

```
    while (start <= end) {
```

```
        int mid = start + (end - start) / 2;
```

```
        if (arr[mid] == LogTarget_ID) {
```

```
            found = true;
```

```
            break;
```

```
        }
```

```
        else if (arr[mid] > LogTarget_ID) {
```

```
            end = mid - 1;
```

```
        }
```

```
        else {
```

```
            start = mid + 1;
```

```
        }
```

```
    }
```

```
}
```

Week2 > BinarySearch.cpp > main()

```
1  #include <iostream>
2  #include <vector>
3
4  int main() {
5      using namespace std;
6
7      int No_of_ID, LogTarget_ID;
8
9      std::cout << "Enter no. of ID : ";
10     std::cin >> No_of_ID;
11
12     std::vector<int> arr(No_of_ID);
13
14     std::cout << "Enter Log IDs : ";
15     for (int i = 0; i < No_of_ID; i++) {
16         std::cin >> arr[i];
17     }
18
19     std::cout << "Enter target Log ID : ";
20     std::cin >> LogTarget_ID;
21
22     int start = 0;
23     int end = No_of_ID - 1;
24     bool found = false;
25
26     while(start <= end){
27         int mid = start + (end - start) / 2;
28
29         if(arr[mid] == LogTarget_ID) {
30             found = true;
31             break;
32         }
33         else if (arr[mid] > LogTarget_ID) {
34             end = mid - 1;
35         }
36         else {
37             start = mid + 1;
```

```
PS C:\MinGW\bin> C:\Users\Lenovo\OneDrive\C++\Week2\LinearSearch.exe
```

```
Enter no. of ID : 5
```

```
Enter Transaction IDs : 1201
```

```
1305
```

```
1450
```

```
1480
```

```
1502
```

```
Enter Suspicious ID : 1450
```

```
Fraud Transaction found
```

```
PS C:\MinGW\bin> █
```



```
int main() {
```

```
std::cout << "Enter no. of ID : ";
```

```
int Fraud_ID, arr[NO_of_ID];
```

```
for (int i=0 ; i < No_of_ID ; i++)
```

3

```
std::cin >> Fraud_ID;
```

- o/p Log found or
 Log Not found.

start mid end = n-1

↓ ↓ ↓

1001	1005	1010	1015	1020	1030	1040
0	1	2	3	4	5	6

$$\text{length} = 7/2 = 3.5 = 3 \rightarrow \text{mid}$$

```
{ found = true;
  break;
```

```
2         end = mid - 1;
```

}

Week2 > LinearSearch.cpp > main()

```
1  #include <iostream>
2
3  int main() {
4      int No_of_ID;
5      std::cout << "Enter no. of ID : ";
6      std::cin >> No_of_ID;
7
8      int Fraud_ID , arr[No_of_ID];
9      bool found=false;
10     std::cout << "Enter Transaction IDs : ";
11     for (int i = 0; i < No_of_ID; i++) {
12         std::cin >> arr[i];
13     }
14
15     std::cout << "Enter Suspicious ID : ";
16     std::cin >> Fraud_ID;
17
18     for(int i = 0 ; i<No_of_ID ; i++)
19     {
20         if(arr[i]==Fraud_ID)
21         {
22             found = true;
23             break;
24         }
25     }
26
27     if(found){
28         std::cout<<"Fraud Transaction found";
29     }
30     else{
31         std::cout<<"Fraud Transaction Not found";
32     }
33     return 0;
34 }
```

Week2 > BinarySearch.cpp > main()

```
35     }
36     else {
37         start = mid + 1;
38     }
39 }
40
41 if(found) {
42     std::cout << "Log Found" ;
43 }
44 else {
45     std::cout << "Log Not Found";
46 }
47
48 return 0;
49 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\MinGW\bin> C:\Users\Lenovo\OneDrive\C++\Week2\BinarySearch.exe
Enter no. of ID : 7
Enter Log IDs : 1001
1005
1010
1015
1020
1030
1040
Enter target Log ID : 1015
Log Found
PS C:\MinGW\bin> |
```

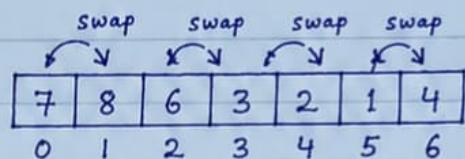


```

if (found) {
    std::cout << "Log Found";
}
else {
    std::cout << "Log Not found";
}
return 0;
}

```

3. Bubble Sort.



```

#include <iostream>
#include <vector>
using namespace std;
int main() {
    int No_of_Employee;
    cout << "Enter no. of Employee : ";
    cin >> No_of_Employee;
    vector<int> arr(No_of_Employee);
    cout << "Enter Salary of Employee : ";
    for (int i=0; i < No_of_Employee; i++) {
        cin >> arr[i];
    }
    for (int i=0; i < No_of_Employee-1; i++) {
        for (int j=0; j < No_of_Employee-i-1; j++) {
            if (arr[j] > arr[j+1]) {
                swap(arr[j], arr[j+1]);
            }
        }
    }
    cout << "Sorted Salaries ";
    for (int i=0; i < No_of_Employee; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}

```



```
36
37 int main() {
38     int No_of_Employee , temp ;
39
40     std::cout << "Enter no. of Employee: ";
41     std::cin >> No_of_Employee;
42
43     std::vector<int> arr(No_of_Employee);
44     std::cout << "Enter Salary of Employee: ";
45     for (int i = 0; i < No_of_Employee; i++) {
46         std::cin >> arr[i];
47     }
48
49     for(int i = 0 ; i < No_of_Employee-1 ; i++)
50     {
51         for (int j = 0; j < No_of_Employee - i - 1; j++) {
52             if(arr[j]>arr[j+1])
53             {
54                 temp = arr[j];
55                 arr[j] = arr[j+1];
56                 arr[j+1] = temp;
57             }
58         }
59     }
60
61     std::cout << "Sorted Salaries: ";
62     for (int i = 0; i < No_of_Employee; i++) {
63         std::cout << arr[i] << " ";
64     }
65     std::cout << std::endl;
66     return 0;
67 }
```

```
PS C:\MinGW\bin> C:\Users\Lenovo\OneDrive\C++\Week2\BubbleSort.exe
```

```
Enter no. of Employee: 5
```

```
Enter Salary of Employee: 45000
```

```
32000
```

```
55000
```

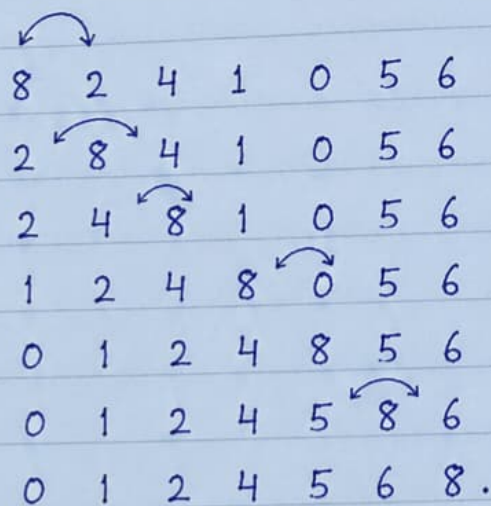
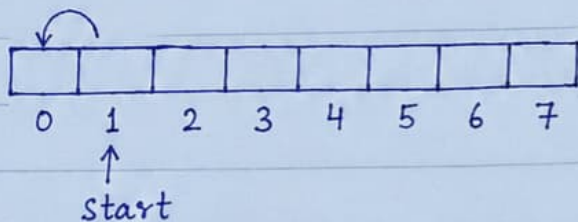
```
28000
```

```
60000
```

```
Sorted Salaries: 28000 32000 45000 55000 60000
```

```
PS C:\MinGW\bin> █
```

4. Insertion Sort.



```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    int n;
    cout << "Enter length : ";
    cin >> n;
    vector<int> arr(n);
    cout << "Enter number : ";
    for (int i=0; i<n; i++) {
        cin >> arr[i];
    }
    for (int i=1; i<n; i++) {
        int key = arr[i];
        int j = i-1;
        while (j >= 0 && arr[j] > key) {
            arr[j+1] = arr[j];
            j--;
        }
        arr[j+1] = key;
    }
    cout << "Sorted : ";
    for (int i=0; i<n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

Week2 > G InsertionSort.cpp > main()

```
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4
5  int main() {
6      int n;
7      cout << "Enter length: ";
8      cin >> n;
9
10     vector<int> arr(n);
11     cout << "Enter numbers: ";
12     for (int i = 0; i < n; i++) {
13         cin >> arr[i];
14     }
15
16     // Insertion Sort
17     for (int i = 1; i < n; i++) {
18         int key = arr[i];
19         int j = i - 1;
20
21         // Shift elements greater than key to the right
22         while (j >= 0 && arr[j] > key) {
23             arr[j + 1] = arr[j];
24             j--;
25         }
26         arr[j + 1] = key;
27     }
28
29     cout << "Sorted: ";
30     for (int i = 0; i < n; i++) {
31         cout << arr[i] << " ";
32     }
33     cout << endl;
34
35     return 0;
36 }
```



```
PS C:\MinGW\bin> C:\Users\Lenovo\OneDrive\C++\Week2\InsertionSort.exe
```

```
Enter length: 5
```

```
Enter numbers: 250
```

```
120
```

```
320
```

```
180
```

```
100
```

```
Sorted: 100 120 180 250 320
```

```
PS C:\MinGW\bin> 
```

• Selection Sort.

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter no. of elements";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " element : ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    for (int i = 0; i < n - 1; i++) {
        int minIndex = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
        int temp = arr[minIndex];
        arr[minIndex] = arr[i];
        arr[i] = temp;
    }
    cout << "Sorted array : ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    return 0;
}
```

Week2 > SelectionSort.cpp > ...

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number of elements: ";
7      cin >> n;
8
9      int arr[n];
10     cout << "Enter " << n << " elements: ";
11     for (int i = 0; i < n; i++) {
12         cin >> arr[i];
13     }
14
15     // Selection Sort logic
16     for (int i = 0; i < n - 1; i++) {
17         int minIndex = i;
18         for (int j = i + 1; j < n; j++) {
19             if (arr[j] < arr[minIndex]) {
20                 minIndex = j;
21             }
22         }
23         // Swap
24         int temp = arr[minIndex];
25         arr[minIndex] = arr[i];
26         arr[i] = temp;
27     }
28
29     cout << "Sorted array: ";
30     for (int i = 0; i < n; i++) {
31         cout << arr[i] << " ";
32     }
33     cout << endl;
34
35     return 0;
36 }
37
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\MinGW\bin> C:\Users\Lenovo\OneDrive\C++\Week2\SelectionSort.exe
```

```
Enter number of elements: 5
```

```
Enter 5 elements: 9
```

```
3
```

```
1
```

```
8
```

```
4
```

```
Sorted array: 1 3 4 8 9
```

```
PS C:\MinGW\bin> █
```